Finite Markov Chain

unichain 解唯一

ergodic unichain 解唯一且 $[P^n] \to e\pi$

P_{ij}^n (n步到) $\to$ recurrent $P_{ij}^n > 0$, aperiodic

ergodic $\Leftrightarrow$ convergence

$\pi_j = \lim_{n\to\infty} P_{ij}^n$ exist 定义 $\qquad \pi = \sum_{i=1}^{n} \pi_i P_{ij} = \pi[P]$ 存在

$[P^n] \to e\pi$

Countable Markov Chain e.g. Birth-Death chain

$f_{ij}(n)$ (n步首达) $\to$ $F_{ij}(n)$ (n步来过)

$\to$ recurrent $F_{jj}(\infty) = 1$, aperiodic

- positive recurrent $\bar{T}_{jj} < \infty$ (生死链 $p < \frac{1}{2}$)
- null recurrent $\bar{T}_{jj} = \infty$ ($p = \frac{1}{2}$)

关联 $\pi_j = \frac{1}{\bar{T}_{jj}}$

irreducible：全连 $\longrightarrow$ ergodic $\Leftrightarrow$ convergence

$\pi_j = \lim_{n\to\infty} P_{ij}^n$ exist 定义 $\qquad \pi_j = \sum_{i=1}^{\infty} \pi_i P_{ij}$ 有解

判定：Foster's theorem

$\Delta V(x) \le c < 0$ 且 $\Delta V(0) + V(0) < \infty$

drift $\Delta V(x) = E(V(x') - V(x) \mid x)$

Stability / Boundness

$\lim_{t\to\infty} E[|X(t)|^p] = 0$ 定义

判定：Comparison theorem

$\Delta V(x) \le -cf(x) + d$ 则 $\limsup_{t\to\infty} \frac{1}{t} \sum_{\tau=0}^{t} E[f(X(\tau)) \mid X(0) = \tau] \le \frac{d}{c}$

Continous Markov Chain e.g. LTI

kernel $P^m(x, A)$ (m步进A)

$\eta_A = \sum_{n=1}^{\infty} I_A(X_n \in A)$ (来A次数) $\to$ recurrent $E(\eta_A \mid X_0 = x) = \infty$

(irreducible：全连) 或全 recurrent 或 n个 transient 部分

判断依据：$\Delta V(x) \le 0$, $\forall x \in C^c$ small set C

则 Harris recurrent $Pr(\eta_A = \infty \mid X_0 = x) = 1$

+ aperiodic = ergodic

$\lim_{n\to\infty} \| P^n(x, \cdot) - \pi(\cdot) \|_{TV} = 0$ 定义

判定：$\Delta V(x) \le -1 + bI_c(x)$

Stability / Boundness

判定：Comparison theorem

$\Delta V(x) \le -cf(x) + d$ 则 $\limsup_{t\to\infty} \frac{1}{t} \sum_{\tau=0}^{t} E[f(X(\tau)) \mid X(0) = \tau] \le \frac{d}{c}$

Large deviation theory

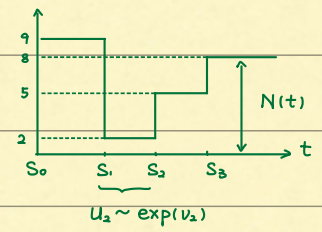
$Pr\{\bigcup_{n=1}^{\infty} (S_n \ge \alpha)\} \le e^{-r^*\alpha}$

$\ln E(e^{rX}) = 0$ 正解 r^*

e.g. M/M/1

Countable Markov Process (= CT)

$X(t)$



e.g. JSQ $\mathcal{L}V(i) = \sum_{j \neq i} q_{ij} [V(j) - V(i)]$

CT-MDP

HJB $\min_u \mathcal{L}W^*$